```
{"add":
  { "torch_operator": "def torch_add(input,
other): …",
    "input_shapes": [
      [[1024, 1024], [1024, 1024]], [[4096,
4096],[4096, 4096]]],
    "triton_operator": "def add_kernel(in_ptr0,
in_ptr1, out_ptr, n_elements, BLOCK_SIZE,): …"
},
… }
```

(a) Operator description

```
{"tools": [
    { "name": "evaluator",
      "signature": "evaluator(src: str, op: str) -> float",
      "purpose": "Compile and execute the generated Triton function triton_<op>
and return its average latency in milliseconds.",
      "inputs": {
        "src": { "type": "string", "description": "…" },
        "op": { "type": "string", "description": "…" } },
      "returns": { "type": "float", "units": "ms", "description": "…" }
}, ...}
```

(b) Tool documentation

```
{"tlx":
  " TLX (Triton Low-level Language Extensions) is a low-level, warp-aware, hardware-near extension of the Triton DSL. It offers
  intrinsics and warp-specialized operations for fine-grained GPU control, hardware-oriented primitives for advanced kernel
  development, and explicit constructs for GPU memory, computation, and asynchronous control flow. TLX is designed for expert users
  pushing Triton closer to the metal.
  - `tlx.async_tasks` and `tlx.async_task`
  ```python
      with tlx.asycn_task(num_warps = 4)
          ...
  ```
  `tlx.async_task(num_warps=4)` defines a warp-specialized asynchronous task that explicitly reserves 4 warps in addition to those
  used by the trunk task. ",
… }
```

(c) Language description

```
{"GPUs": [
    { "name": "H100_SXM5",
      "process": "4nm",
      "transistor_count": "80 billion",
      "memory": {
        "type": "HBM3",
        "capacity": "80 GB",
        "bandwidth": "3000 GB/s"
      },
      "vector_performance": {
        "fp64": "30 TFLOPS", …},
      "tensor_performance": {
        "fp64": "60 TFLOPS", …},
      "SM_num": "132",
      "GPC_num": "7",
      "on_chip_mem": {
        "LLC_size": "50MB",…},
      "compute_capability": {
        "thread_per_warp": 32, …},
        "power_consumption": "700W"
    },
  …]
}
```

(d) Hardware specification